Microprocessors And Microcontrollers

Assignment 4- Week 4

TYPE OF QUESTION: MCQ

Number of questions: 15 **Total mark: 15 X 1 = 15**

QUESTION 1:

Correct Answer: b

Detailed Solution:

QUESTION 2:

Correct Answer: c

Detailed Solution:

The **Interrupt Vector Table (IVT) of the 8085 microprocessor** is a fixed memory location range where the addresses of the interrupt service routines (ISR) are stored.

Interrupt Vector Table (IVT) Range in 8085

- The **8085 processor** has **vectored interrupts** that jump to predefined memory locations.
- Each vectored interrupt has a **fixed 8-bit address**, meaning it must be within the first **256 bytes** of memory.
- These addresses range from **0000H to 00FFH**.

Breakdown of Interrupt Vector Addresses:

Interrupt	Vector Address
TRAP	0024H
RST 7.5	003CH
RST 6.5	0034H
RST 5.5	002CH

Interrupt	Vector Address
INTR (if vectored externally)	User-defined
RST 0	0000H
RST 1	0008H
RST 2	0010H
RST 3	0018H
RST 4	0020H
RST 5	0028H
RST 6	0030H
RST 7	0038H

QUESTION 3:

Correct Answer: b

Detailed Solution:

When RST 7 instruction encountered corresponding ISR starting at memory location 7*8 = $(56)_{10} = (38)_{16}$ is executed.

QUESTION 4:

Correct Answer: c

Detailed Solution:

QUESTION 5:

INTR must remain active for how many T-states?

A) 16.5 T-states
B)17.5 T-states
C) 18.5 T-states
D) 19.5 T-states

Correct Answer: B

Detailed Solution:

QUESTION 6:

Multiply 03H and B2H (two 8-bit numbers) stored in memory locations 2200H and 2201H by repetitive addition and store the result in memory locations 2300H and 2301H. The output in 2300H and 2301H is

a) 04H and 16H

b) 16H and 02H

c) 03H and 16H

d) 16H and 03H

Correct Answer: b

Detailed Solution: (2200H) = 03H ; (2201H) = B2H

Result = B2H + B2H + B2H = 216H ; (2300H) = 16H ; (2301H) = 02H

QUESTION 7:

Correct Answer: a

Detailed Solution:

QUESTION 8:

Correct Answer: b

Detailed Solution:

QUESTION 9:

Correct Answer: c

Detailed Solution:

QUESTION 10:

Correct Answer: b

Detailed Solution:

QUESTION 11:

Correct Answer: b

Detailed Solution:

QUESTION 12:

Correct Answer: c

Detailed Solution:

QUESTION 13:

Correct Answer: a

Detailed Solution:

Bit 4 of the accumulator in the SIM instruction allows explicitly

resetting the RST 7.5 memory

QUESTION 14:

Correct Answer: b

Detailed Solution:

QUESTION 15:

How many 8-bit characters can be transmitted per second over a 9600 baud serial communication link using asynchronous mode of transmission with one start bit, eight data bits, two stop bits, and one parity bit?

a) 600

b) 800

c) 1000

d) 1200

Correct Answer: b

Detailed Solution: "9600 baud" means that the serial port is capable of transferring a maximum of 9600 bits per second.

Total Data To send = 1 bit(start) + 8 bits (char size) + 1 bit (Parity) + 2 bits (Stop) = 12 bits.

Number of 8-bit characters that can be transmitted per second = 9600/12 = 800.

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